26F Stat TA Session W13

0. Preparation for TA Session

- Download the sample code and and attached dataset `china` , `textbooks` , and `ncbirths` (click here)
- Import the datasets into your environment and name them as `china` , `book` , and `birth` , respectively
  - `china` records the weekly time spent on child care in China
  - `book` contains prices of textbooks for various courses at UCLA bookstore and Amazon
  - `birth` is a 2004 dataset from North Carolina on births, including mothers' habits and practices, with 1,000 cases

1. Student's t-test `t.test()`

- Similar to Chi-Squared test, t-test is another useful and thus common hypothesis testing
- There are three situations to use t -test!

One-Sample t-test

- **When to use:** test if the population mean equals a target μ

```
# syntax
t.test(vec, mu,
             alternative, conf.level)
# alternative hypothesis defaults "two.sided"
# you can also use "less" or "greater"

# example
china <- china %>%
  filter(child_care >= 0)

china_male <- china %>%
  filter(gender == 1)

t.test(china_male$child_care,
       mu = 25,
       conf.level = 0.95)

# output
    One Sample t-test

data:  china_male$child_care
```

```
t = -9.2583, df = 357, p-value < 2.2e-16

alternative hypothesis: true mean is not equal to 25

95 percent confidence interval:
 13.17385 17.31777

sample estimates:
mean of x
 15.24581

# example
t.test(china$child_care,
       mu = 25,
       alternative = "less",
       conf.level = 0.95)

# output
    One Sample t-test

data:  china$child_care
t = -0.73041, df = 1021, p-value = 0.2327

alternative hypothesis: true mean is less than 25

95 percent confidence interval:
     -Inf 25.79633

sample estimates:
mean of x
 24.36497
```

Paired Sample t-test

- **When to use:**
  - pre/post on the same subjects or two methods measured on the same item
  - test if the mean difference equals μ (usually we want to check if the difference is 0)

```
# syntax
t.test(vec_1, vec_2,
       paired = TRUE, conf.level)

# example
t.test(book$ucla_new,
```

```
        book$amaz_new,
        paired = TRUE,
        coef.level = 0.95)

# output
    Paired t-test

data:  book$ucla_new and book$amaz_new
t = 7.6488, df = 72, p-value = 6.928e-11

alternative hypothesis: true mean difference is not equal to 0

95 percent confidence interval:
  9.435636 16.087652

sample estimates:
mean difference
       12.76164
```

Two Independent Sample t-test

- **When to use:** compare the means of two distinct groups (e.g., treatment group vs control group; smokers vs non-smokers)

```
# syntax
t.test(data$ValueVar ~ data$GroupVar,
       conf.level)

# example
# Do newborns' weights differ btw smoking & nonsmoking mothers?

t.test(birth$weight ~ birth$habit,
       conf.level = 0.95)

# output
    Welch Two Sample t-test

data:  birth$weight by birth$habit
t = 2.359, df = 171.32, p-value = 0.01945

alternative hypothesis: true difference in means between group nonsmoker and
group smoker is not equal to 0

95 percent confidence interval:
 0.05151165 0.57957328
```

```
sample estimates:
mean in group nonsmoker    mean in group smoker
               7.144273                6.828730
```

▼ Q. What's the difference between Chi-Squared test and t-test?

Ans.

- A t-test is used to determine if there is a significant difference between **the means of numerical data.**
- A Chi-Squared test is for **categorical data** instead, and is used to determine if there is a significant **association between variables** (Independence) or if the **observed frequencies differ from the expected frequencies** (Goodness-of-Fit)

Exercises

W13 Exercises